Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/21/2023
PI Name:	DeChurch, Leslie Ph.D.		
Project Title:	Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking,& Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Previously at Georgia Institute of Technology until July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	11/09/2017	**End Date:**	03/31/2024
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	12	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2024 per A. Beitman/NASA JSC (Ed., 10/5/23) NOTE: End date changed to 11/08/2023 per A. Beitman/NASA JSC (Ed., 11/3/22) NOTE: End date changed to 11/08/2022 per NSSC information. (Ed. 10/29/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Contractor, Noshir Ph.D. (Northwestern University)		
Grant/Contract No.:	80NSSC18K0276		
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Task Description:	NOTE: Continuation of "Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking, & Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews," grant NNX15AK73G, due to Principal Investigator move to Northwestern University from Georgia Tech in Fall 2016. We are at the dawn of a new era of human space exploration. Moving beyond low Earth orbit and the relative safety of the International Space Station (ISS) toward near-Earth asteroids and Mars presents previously unimaginable opportunities as well as organizational challenges. One significant challenge is the complexity of the operating environment within which astronauts will work. This complexity will place enormous demands on astronauts, and research is needed that develops concrete countermeasures to mitigate the risks stemming from performance decrements due to inadequate cooperation, coordination, communication, and psychosocial adaptation within a team. Astronauts will push the bounds of human cognitive and social functioning as they multitask across tasks, teams, and tools working toward personal, team, and system goals. This multidisciplinary research project is designed to help them do just that. This investigation into team task switching leverages: (1) agent-based models to understand how task switching behavior and performance-related switching costs occur based on dynamic interplay between independent and interdependent tasks, (2) laboratory experiments conducted in two multiteam systems laboratories (one at Georgia Tech, the other at Northwestern) to test the theoretical model derived from agent-based models, (3) unobtrusive data collection strategies to capture information on the tasks and complex social network structures of ISS crew members, (4) design of interventions to enable adaptive team task switches using virtual experiments, (5) validation of intervention strategies using one of NASA's space analogs, and (6) development of a dashboard decision aid to anticipate and pre-empt dysfunctional task switching. An innovative feature of our investigation is the use of a multidimensional network approach to characterize and model the switches between tasks, tools, teams, and multi-team systems. The combined outputs of this proposed multi-disciplinary project speak directly to NASA Human Research Program's (HRP's) identified team gaps for autonomous, long duration, and/or long distance exploration missions associated with the need to identify (1) key threats to the team (Team Gap 1), (2) countermeasures to support team function (Team Gap 3), (3) validated methods that can be used to maintain team function (Team Gap 5), and (4) psychological and psychosocial factors, measures, and combinations thereof that can be used to compose highly effective crews (Team Gap 8). [Ed. note Sept 2020: HRP Gaps have changed since original proposal--see Human Research Roadmap https://]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous models of task switching revolve around task and individual factors that affect work efficiency as individuals move from task to task throughout their workday. However, individuals not only accomplish tasks individually using one technological tool, they do so as members of multiple teams using a variety of technologies. Our research on team task switching examines factors that affect the decision and ability of individuals to switch between tasks with varying levels of interdependence, as well as discuss the mediators and moderators that determine adaptive switching in today's complicated work environments. This model affords a comprehensive understanding of how individuals adapt to dynamic, environmentally-triggered performance demands requiring them to change tasks, teams, and technologies, and shift back and forth between personal, team, and system goals. Our research will be useful for investigating task switching in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN - the European Council for Nuclear Research), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems.
Task Progress:	Over the course of this research program, we (1) explore issues that can arise when astronauts switch between tasks, teams, and technologies within the complex, multiteam systems that characterize operations on the International Space Station (ISS) and during Long Duration Space Exploration (LDSE), as well as (2) investigate the effects of astronaut task, team, and technology switches on individual-, team-, and system-level performance. In particular, astronauts work individually as well as within and across different teams to accomplish their work. They also complete a variety of tasks and engage a variety of technologies in the process. Shifting across these tasks, teams, and tools holds the potential for "switch costs" in the form of process loss and performance decrements. Aspects of the various tasks, teams, and technologies make switching to/from each more/less seamless. We explore the implications of effective/efficient switching. In this review period (year 8), we have cleaned and analyzed the data collected in SIRIUS-21. For this mission, we created new task prompts and increased the number of team tasks to be completed. In conjunction with the NEK SIRIUS-21 mission, we assembled two "twin teams" composed of undergraduate students to participate in the same team and multiteam exercises as the SIRIUS-21 crew outside of isolation. The undergraduate students were recruited from Northwestern University. The qualifications for recruitment were designed to mirror those of the SIRIUS-21 mission. Specifically, we looked for students with science, technology, engineering, and mathematics backgrounds and/or military experience and enthusiasm for the project. Potential participants were asked to fill out a recruitment survey that asked questions about their background and availability, and questions designed to gauge interest. These questions included: "Why do you want to participate?", "What makes you a good team member?", and "Would you consider going to Mars?" After identifying those whose backgrounds matched our criteria and meeting with the students to explain the project's demands, we composed two teams: TWIN-BLACK and TWIN-SILVER. Each team was constructed to have the most diversity among backgrounds, gender, race, and nationality. Each team had seven members to correspond to each of the SIRIUS-21 crew roles plus one backup crewmember. While in isolation, the SIRIUS-21 crew participated in various team tasks designed to evaluate different aspects of team performance. These tasks include a weekly team task battery that contains tasks related to problem-solving, ethical decision-making, and creativity (Larson et al., 2019), and a group decision-making task entitled Hidden Profiles (Stasser & Stewart, 1992) given five times throughout the eight months. Additionally, the non-isolated twin teams met at least once per week, either in person at a research laboratory at Northwestern University or online using Zoom, depending on the Northwestern University academic calendar, to conduct the same team tasks as the SIRIUS-21 crew for the duration of the eight-month mission. The twin teams would meet once a week for an hour to take part in the team task battery. The twin teams also met for five additional hours spread throughout the 8 months to complete the hidden profile decision-making task.

Every two weeks, the SIRIUS-21 crew participated in a multiteam task, Project RED, synchronously with one of the twin teams or the other, alternating each time. The Project RED Design task is a computer-based task requiring a 12-person multiteam system to collaborate to build a hypothetical well to sustain a Mars population (Lungeanu, Mesmer-Magnus, et al., 2022). The participants are assigned to one of the four disciplinary teams (i.e., geology, robotics, human factors, and engineering). Each disciplinary team contained at least one SIRIUS-21 crewmember. Participants performed the task for 45 minutes.

First, applying social network theory and methods, we explored three topological aspects of team social relations found to predict their capacity to perform effectively. These include (1) the level of interconnectedness among the crew, (2) the degree to which the crew shows hierarchy, and variation in status, position, or power, and (3) the extent to which the crew shows subgrouping among members. Results showed that there are substantial differences between the SIRIUS-21 crew and control teams. The motivation ties were strongest in SIRIUS-21 as compared to non-isolated controls. At the same time, SIRIUS-21 experienced the most hindrance. Importantly, the loss of one SIRIUS-21 crewmember on mission day 32 was associated with degraded crew networks, creating subgroups that persisted for the duration of the mission.

Second, to understand how members of the spaceflight multiteam system (SMTS) manage relationships within and across multiple types of boundaries, defined by differences in distance (i.e., space vs. Earth) and goals (i.e., prevention vs. promotion), we studied the formation of affective, behavioral, and cognitive ties among the 12-member space exploration SMTS who participated in the Project RED multiteam task. Interactions took place by text-based interface within the Project RED software. The software captured each member's actions and interactions and launched periodic network surveys. Network surveys elicited affective, behavioral, and cognitive relations. Affective ties were assessed with the prompts “Who did you rely on for leadership?” and “Who kept your task force motivated?” Behavioral ties were assessed with the prompts “Who do you work effectively with?” and “Who worked hard for the task force?” Cognitive ties were assessed with the prompts “Who was a valuable source of information?” and “Who was instrumental in helping the task force achieve its goals?” The task was completed 17 times during the 8-month isolation mission. We used the E-I index to examine the relative internal and external focus of SIRIUS-21 and MAG. Results show that relations formed within and among teams were strongly shaped by location-based differences, with a substantial inward focus: SMTS members formed a greater proportion of relationships with those who are proximal, than with those with whom they work remotely. Results also show that SMTSs members were more likely to form external ties with those on teams with the opposite goal focus.

Bibliography Type:	Description: (Last Updated: 11/17/2025)
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